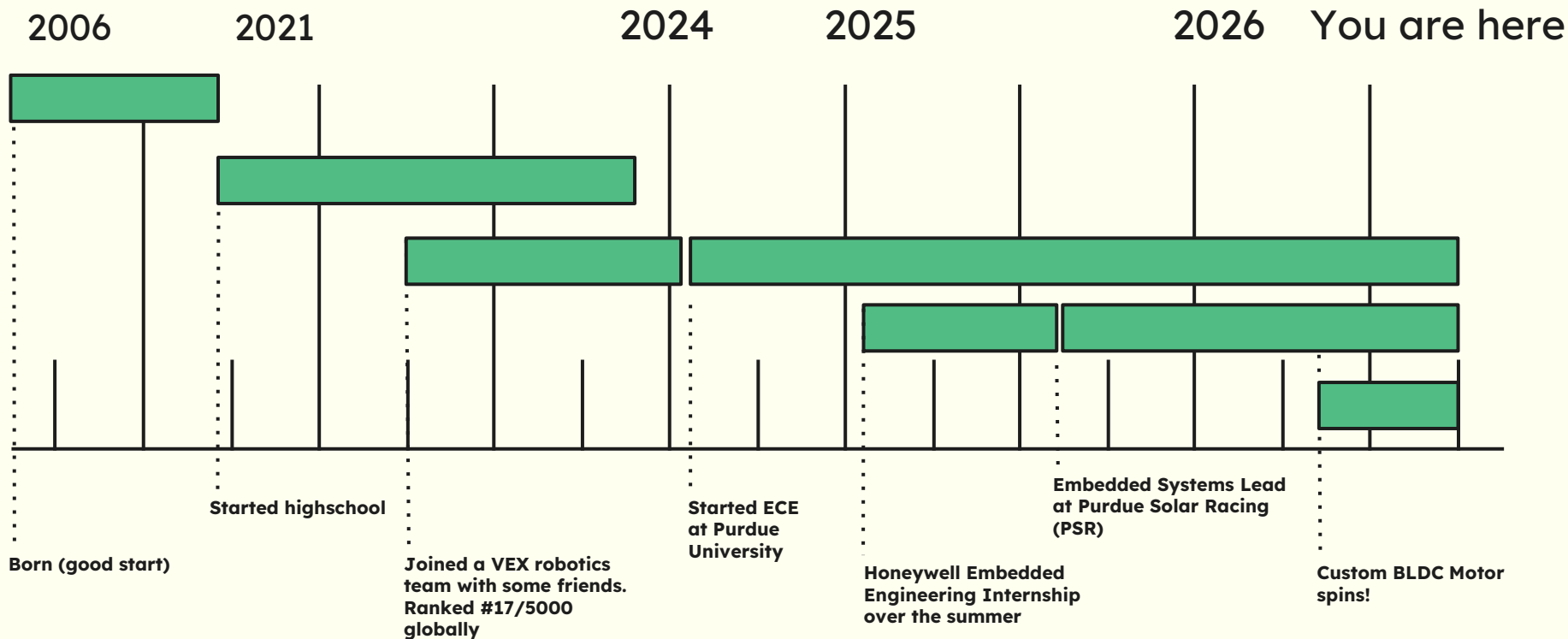


Aditya Verma
Purdue EE 2028

A Collection of Cool Things I've Made

My Career

2



Brushless DC Motor + Controller

3

Designed 36-slot, 42-pole **8010 outrunner BLDC**, 3D printed housing

Hand-wound 3 phases (wye terminated) with 6 parallel, 26 gauge enameled wire for high torque-per-amp, minimal cogging, low-kV

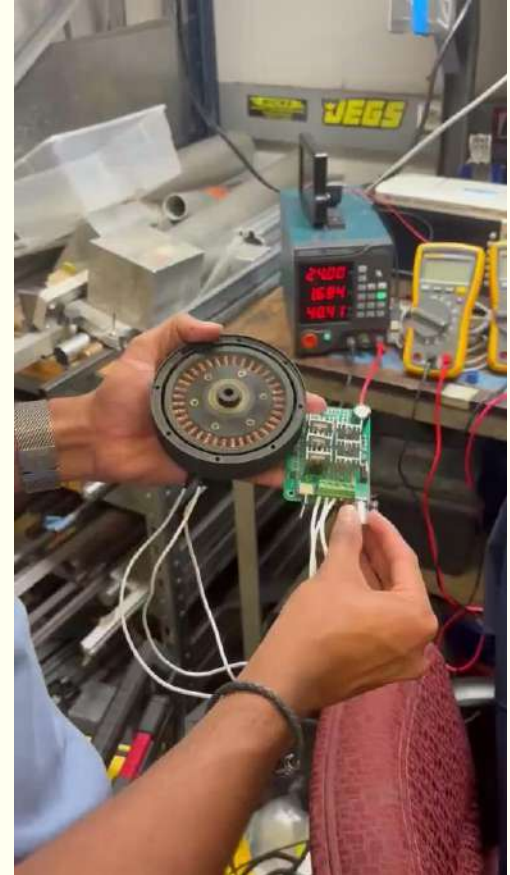
Building **FOC-capable controller PCB**: 10 A/phase, 2 oz copper, Kelvin-sensed shunts, independent on/off gate edge tuning

STM32 integrated with gate driver, SPI for **hall effect** (positional feedback based on magnet), **UART to USB bridge** for PC connection

Tuned system for **2-97% duty cycle at 20 kHz switching**, optimizing current ripple, switching losses, and bootstrap cap sizing

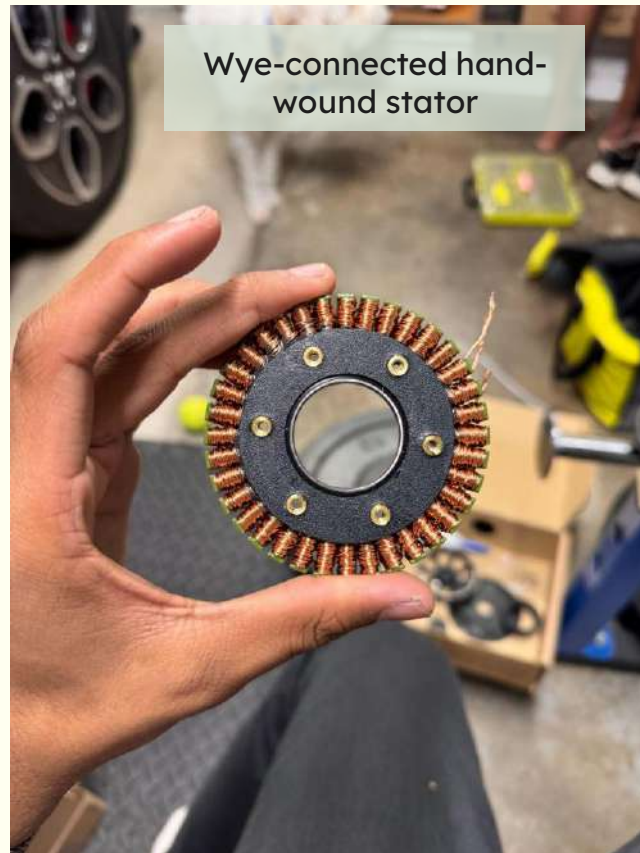
Integrated **reverse-polarity** and **over-voltage** protection with regenerative braking capability for 40 V LiPo operation

4-layer PCB layout focused on high current capability, **analog signal integrity** (Kelvin-sensed shunts), and **thermal dissipation**



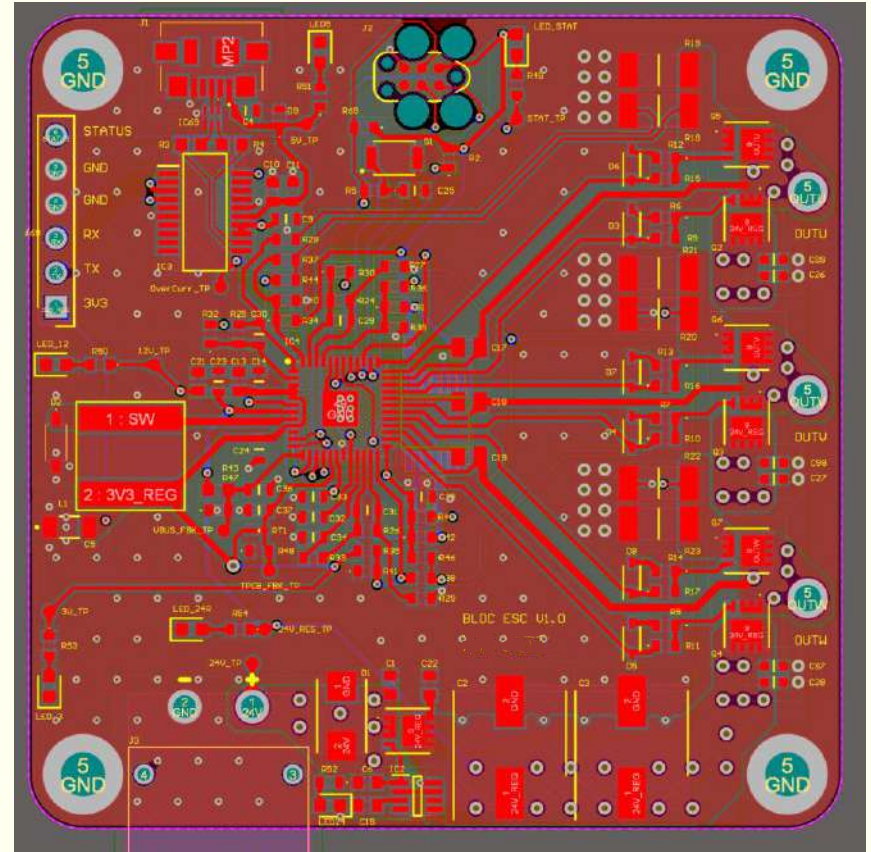
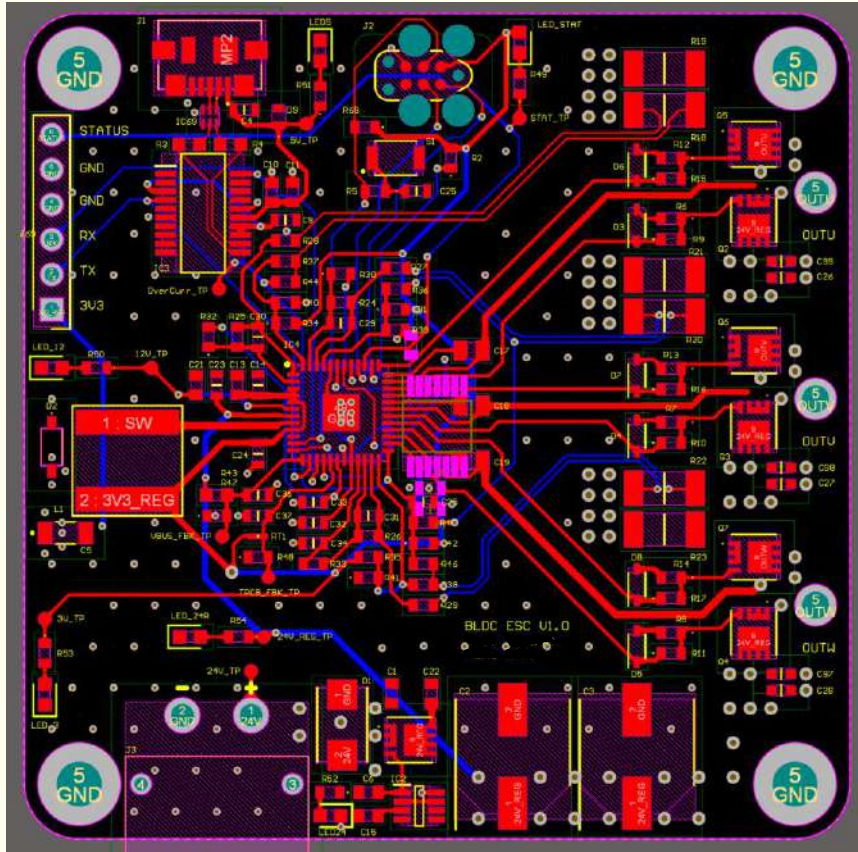
Brushless DC Motor + Controller

4

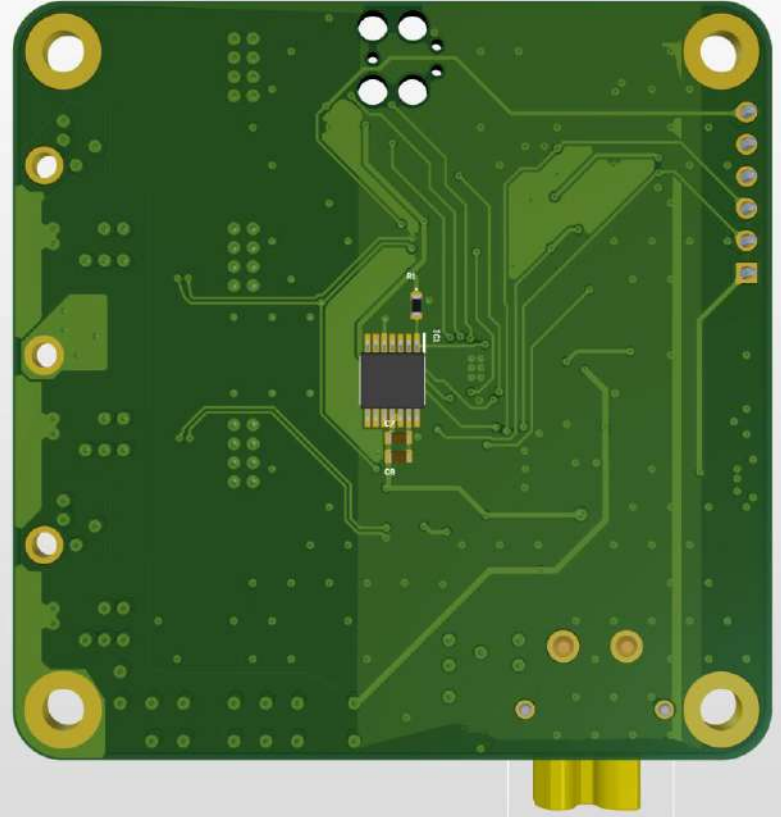
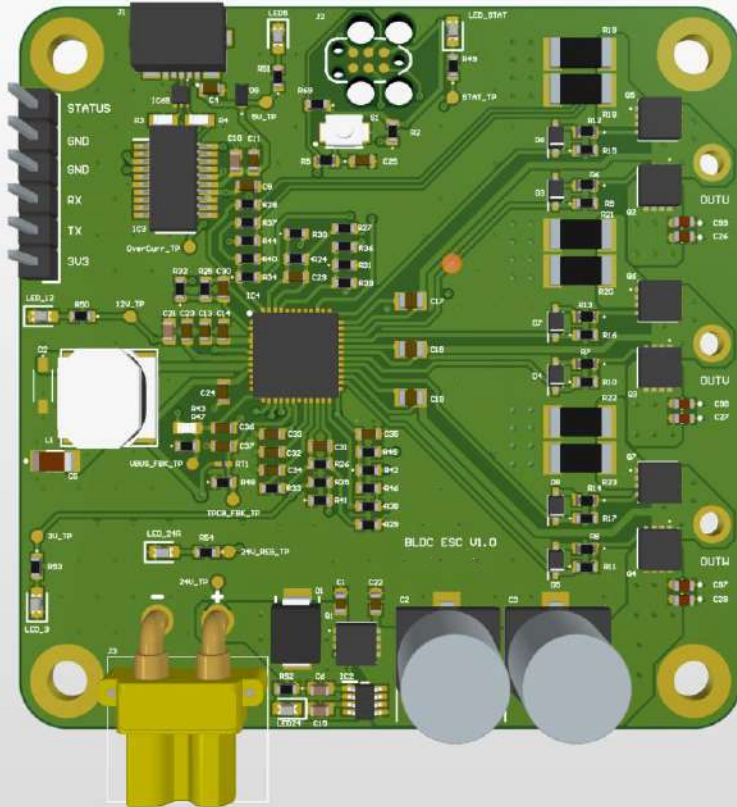


BLDC Controller Layout

5



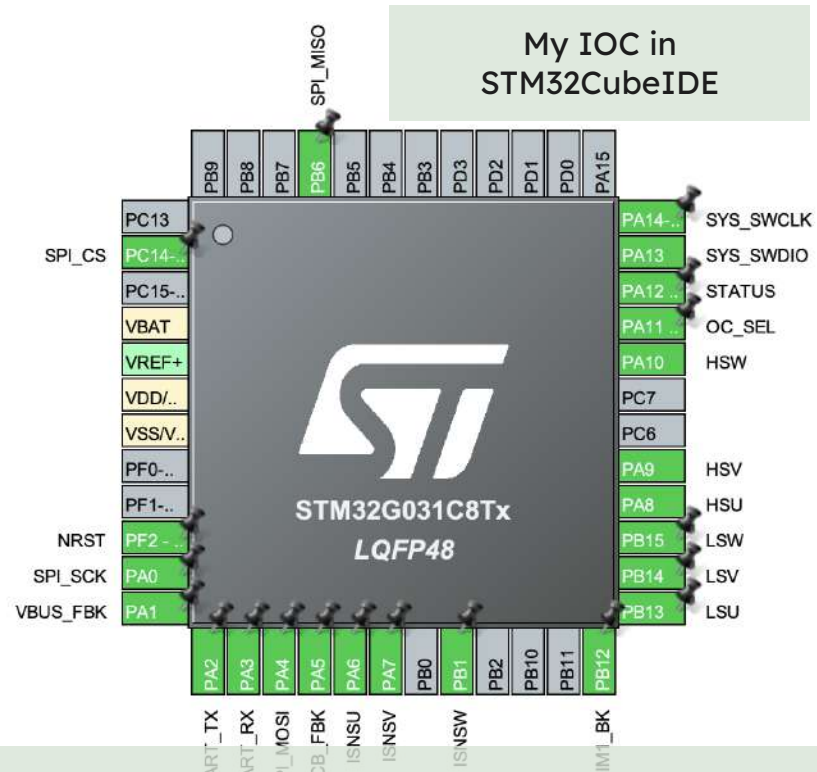
BLDC Controller Layout



BLDC Controller Soldered + Testing

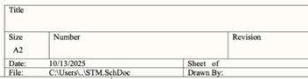
7

Soldered V1 of board

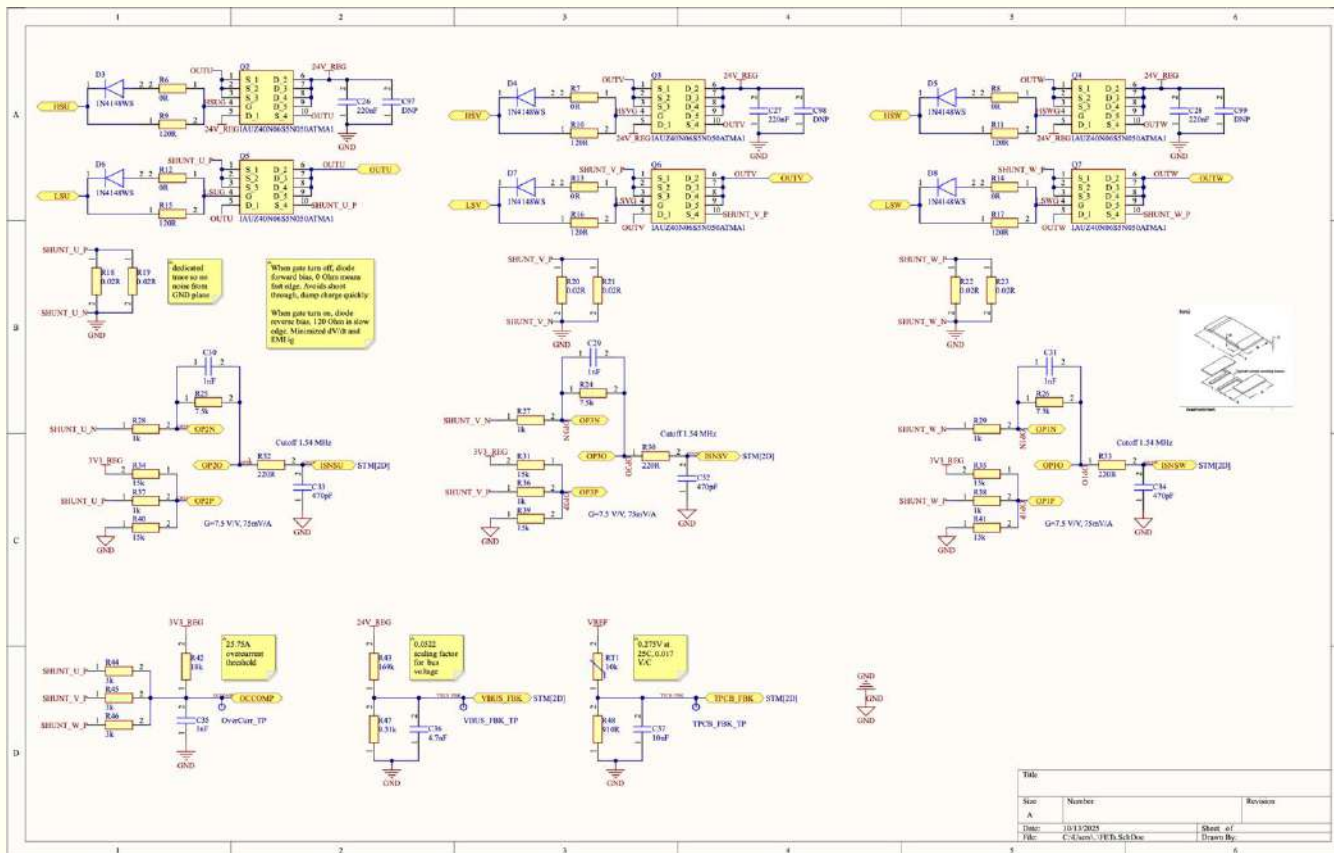


Integrated timer for 20kHz PWM generation with ADC sampling in middle of 50us PWM cycles

8

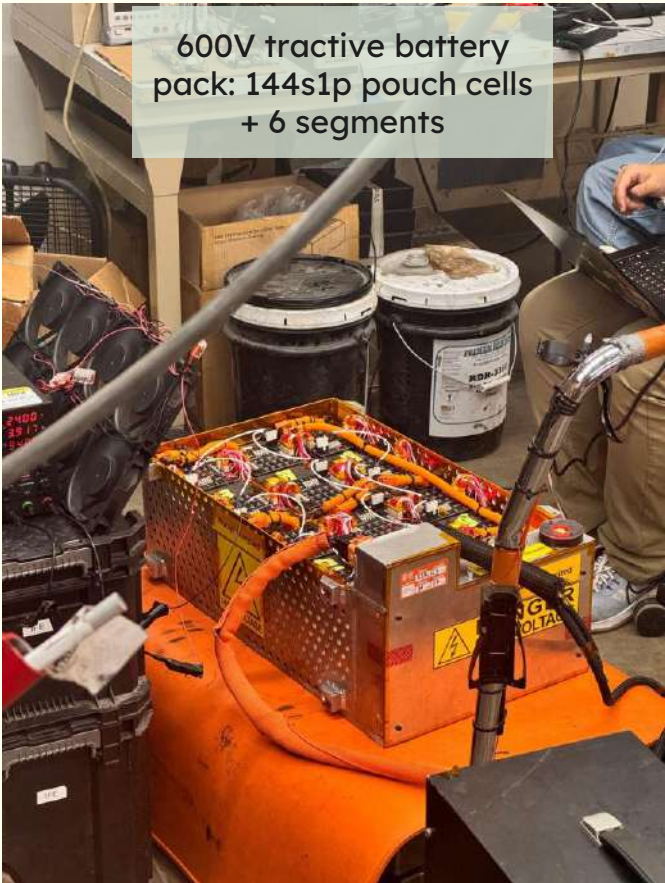


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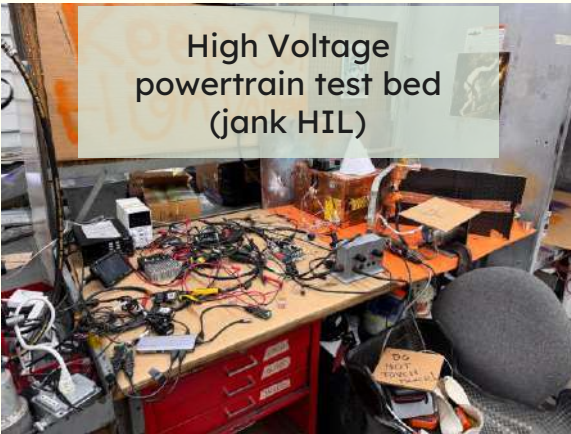


PSR Battery Management System

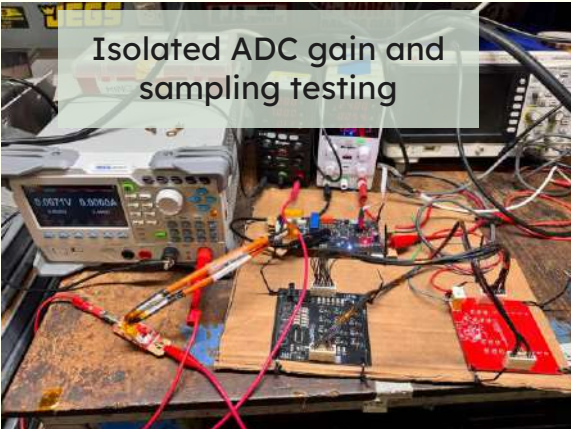
10



600V tractive battery
pack: 144s1p pouch cells
+ 6 segments



High Voltage
powertrain test bed
(jank HIL)



Isolated ADC gain and
sampling testing

Lead architecture, design and validation of **600 V battery management system (BMS)** and **high-voltage power electronics**

Manufactured, validated, debugged custom PCBs: Safety (vehicle state machine, SPI, CAN), steering wheel (I^2C)

Optimized and tuned performance of **analog filters and peripherals** with oscilloscope, waveform gen, load tester, and LTspice

BMS Primary PCB

11

Manages **fault detection**, **precharge**, **isolated ADC**, communication with vehicle control board

CAN 1 (car-critical) and CAN 2 (logging) with transceivers to STM

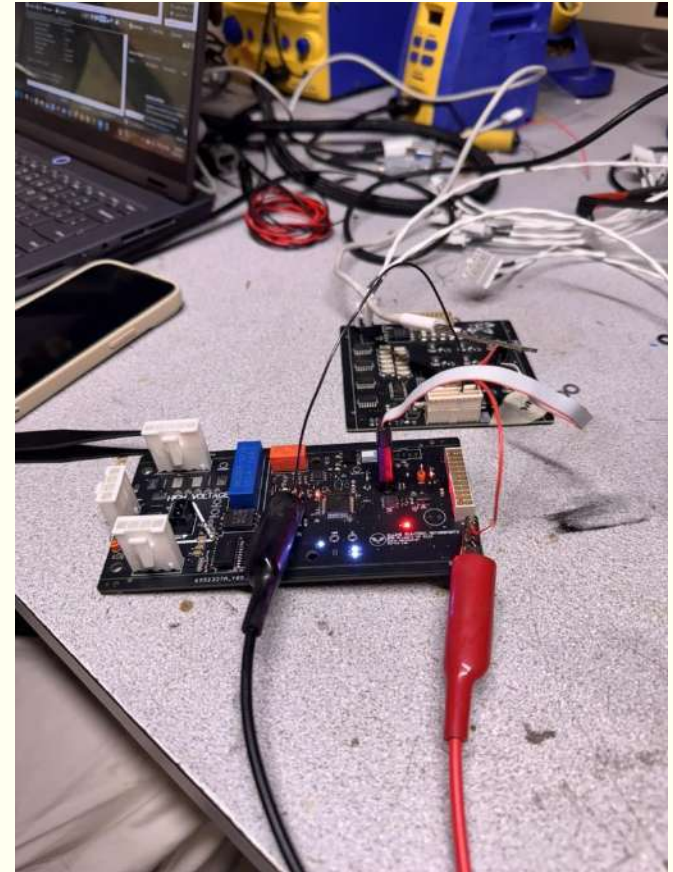
SPI for isoADC to **measure pack status** (for SOC, SOH, SOP)

SPI for flash storage (data logging)

SPI for **communication to secondaries** (isolated through AD proprietary isoSPI with external transformer)

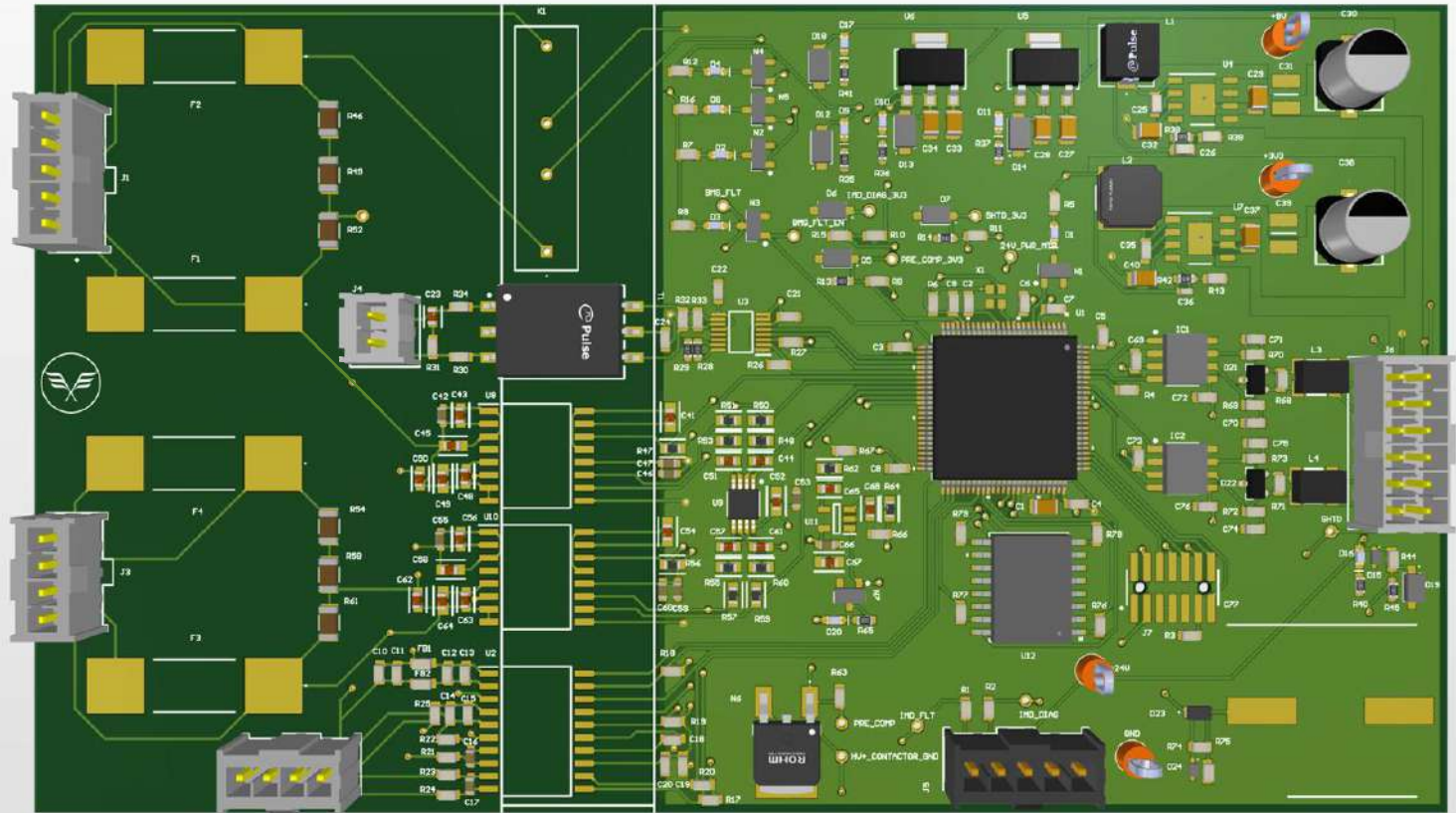
24V - 8V Buck, 8V - 5V LDO, 8V - 3V3 LDO

Layout for **HV isolation**, creepage and clearance



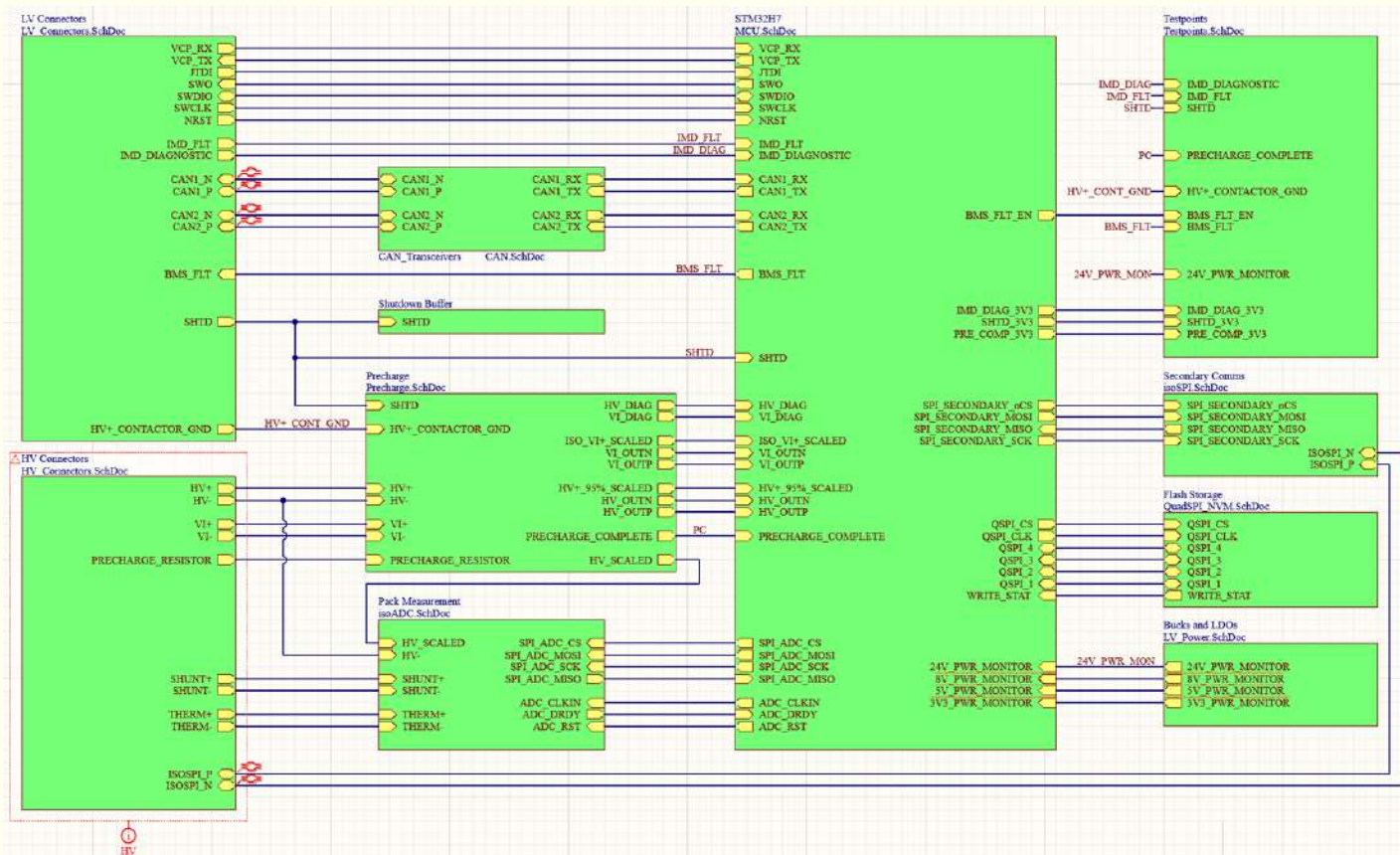
BMS Primary PCB - Layout

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BMS Primary PCB - Schematics

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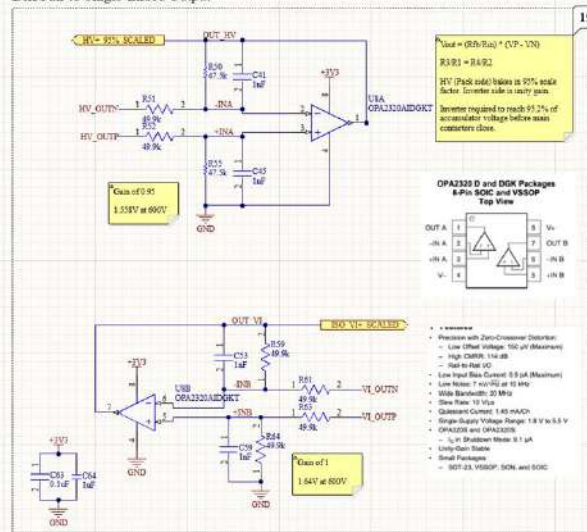


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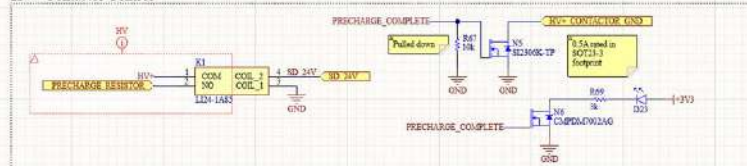
Title		
Size	Number	Review

15

Diff Pair to Single-Ended Output

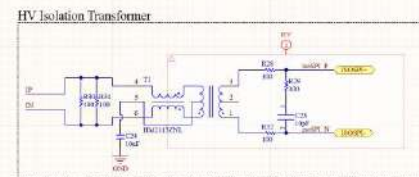
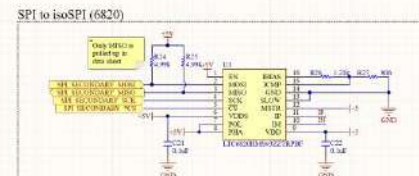
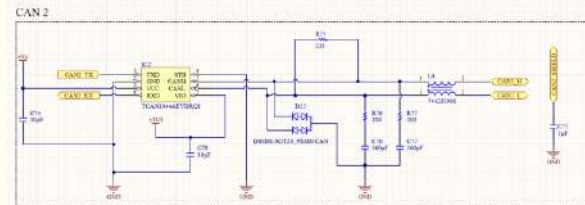
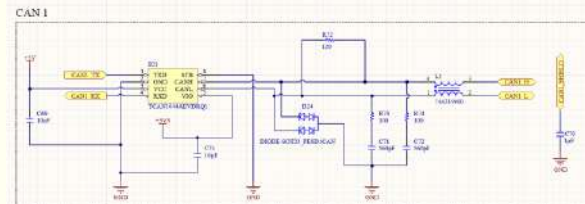
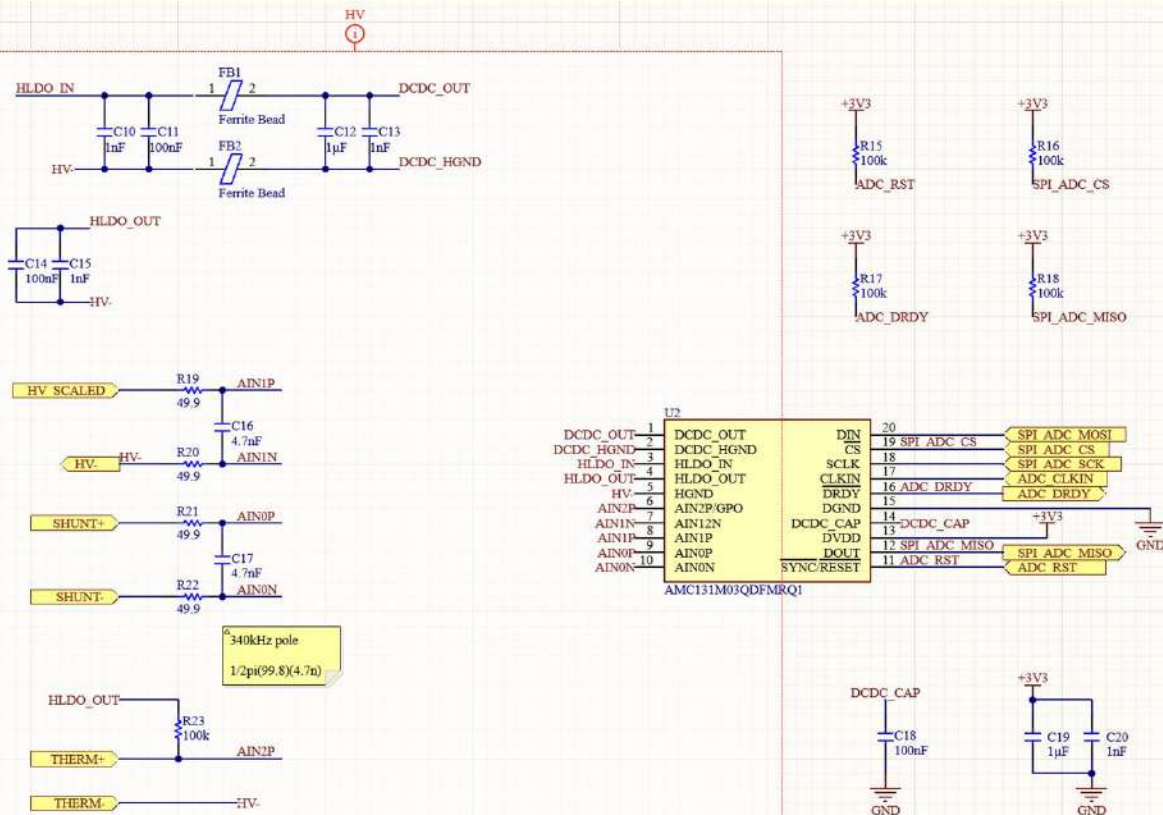


HV+ Contactor Control



BMS Primary PCB - Schematics

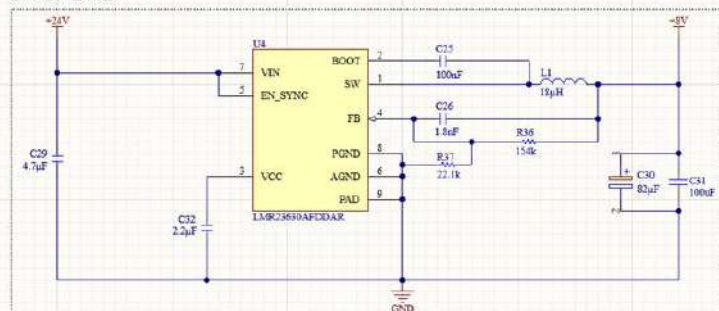
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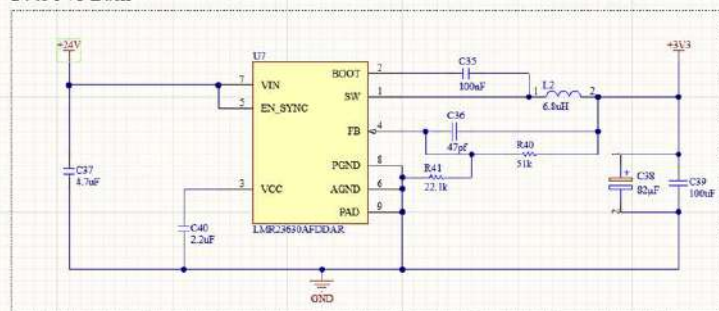
BMS Primary PCB - Schematics

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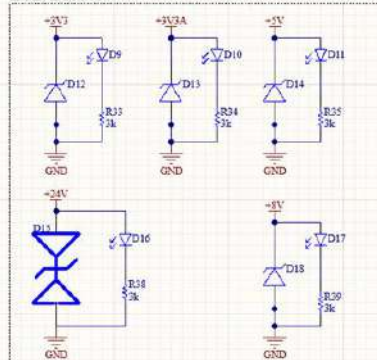
24 - 8V Buck



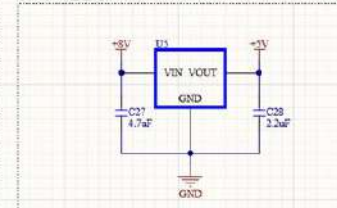
24 to 3V3 Buck



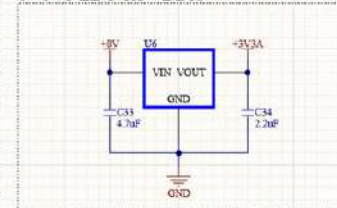
ESD Diodes



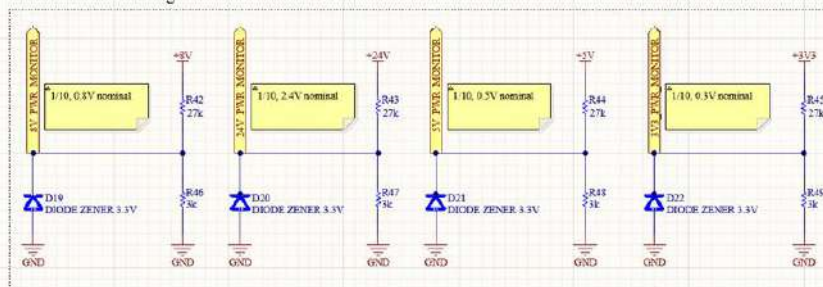
8 - 5V LDO



8 - 3V3 LDO for VDDA



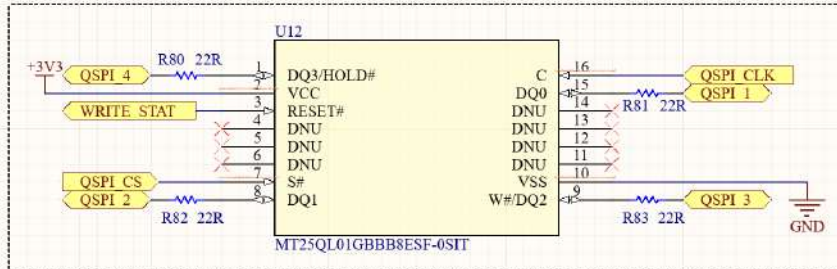
LV Power Monitoring



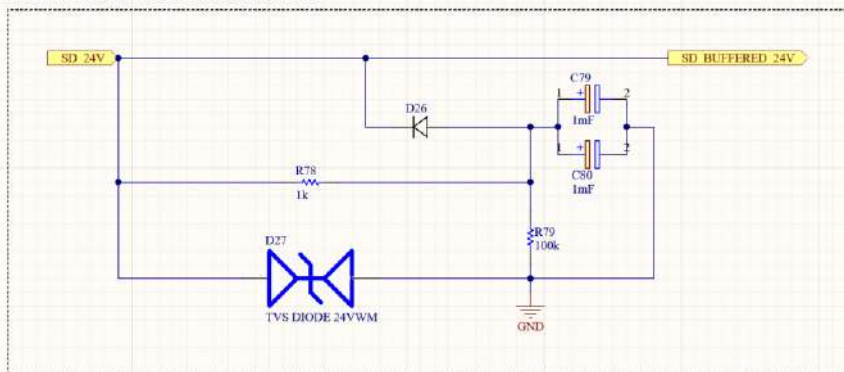
BMS Primary PCB - Schematics

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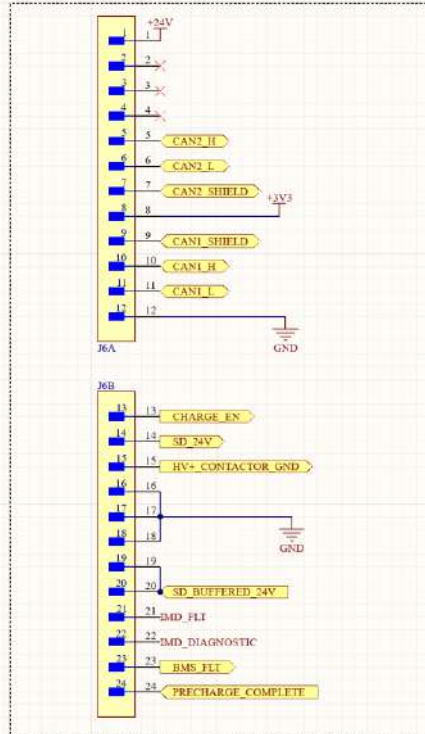
1 Gbit NOR Flash



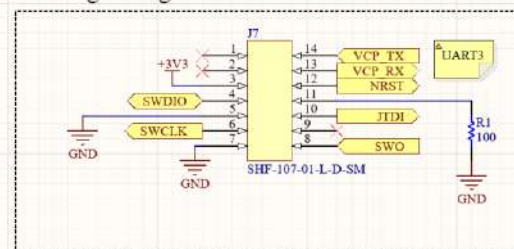
150ms Shutdown Contactor Buffer



24 Pin I/V Samtec



Flashing/Debug



Pack-Side and Precharge Resistor



Shunt



Inverter-Side



isoSPI



BMS Secondary PCB

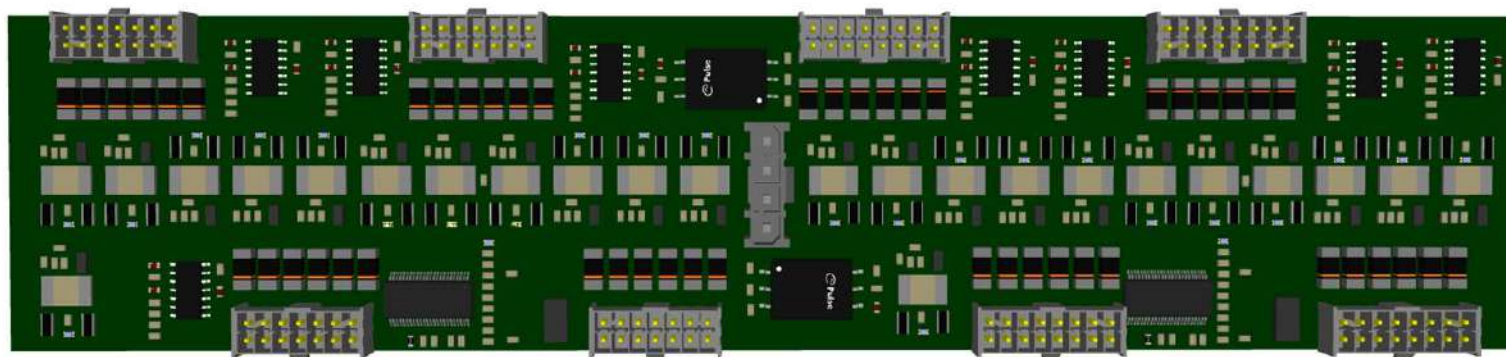
19

Measures **144 individual cell voltages** and temperatures with fused connections
Powered off cells (isolated from LV system)

isoSPI daisy chain for live cell telemetry between 6 secondary PCBs and primary PCB

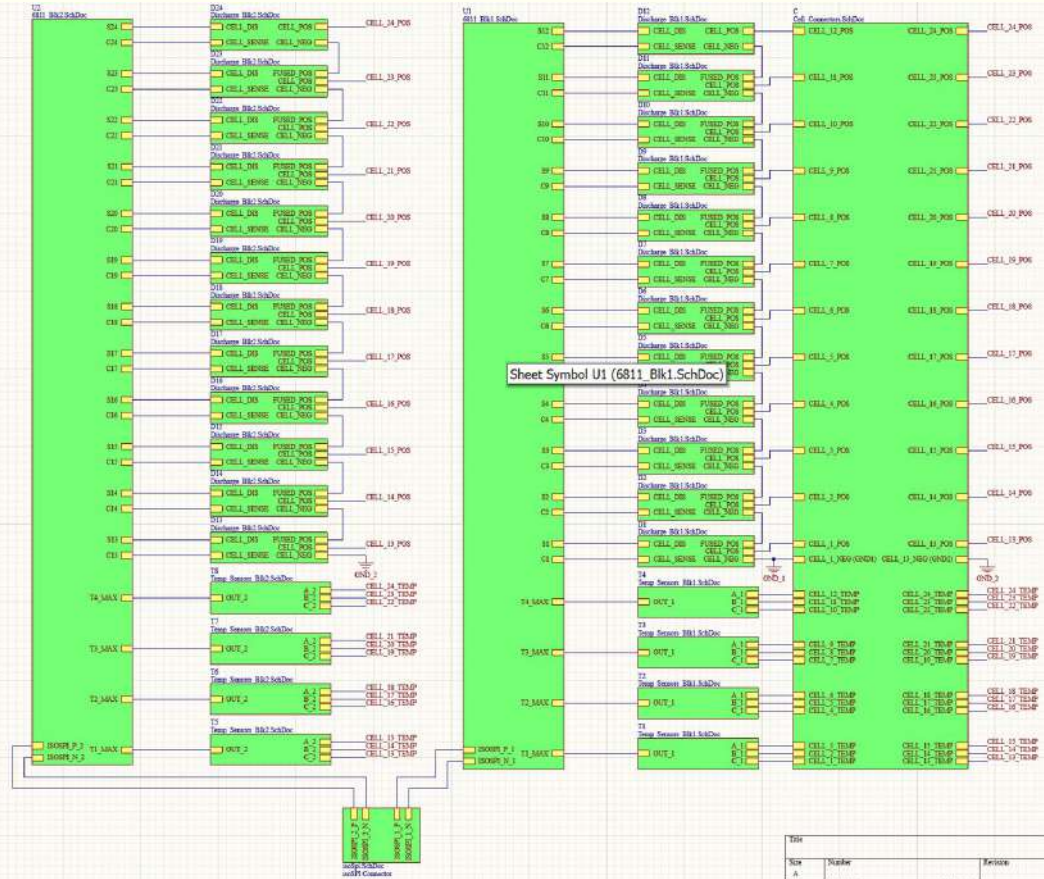
Op-amp network to measure **hottest-of-3 cells** (limited analog inputs to 6811 IC)

New Design: PCB as segment lid, 2x faster cell balancing with thermal jumpers for dissipation



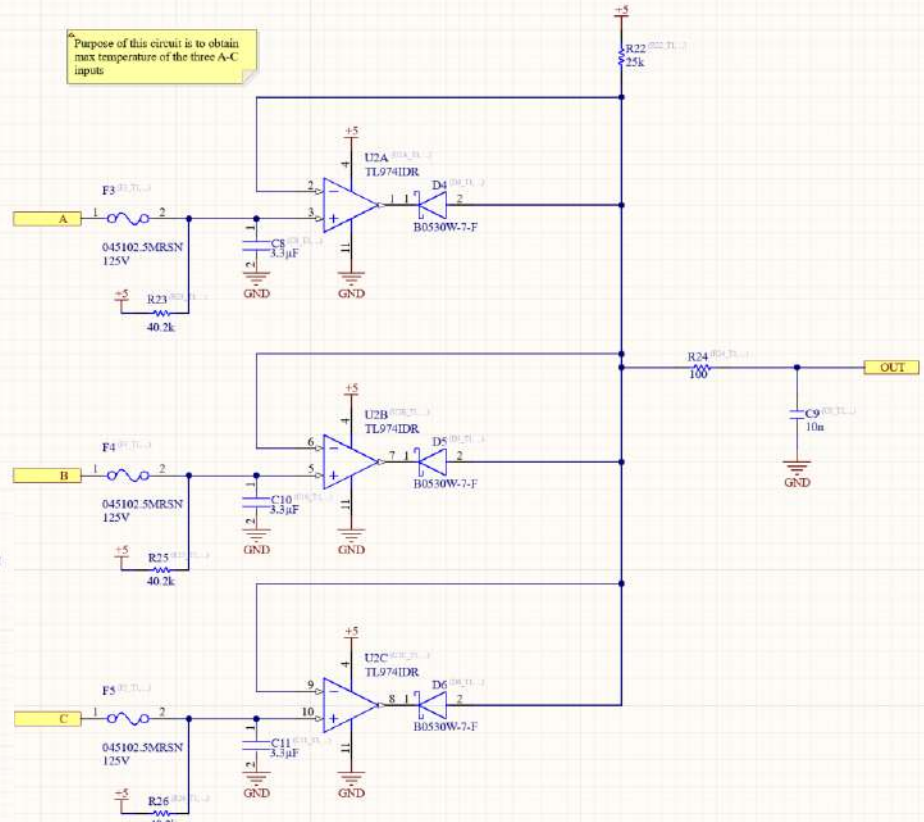
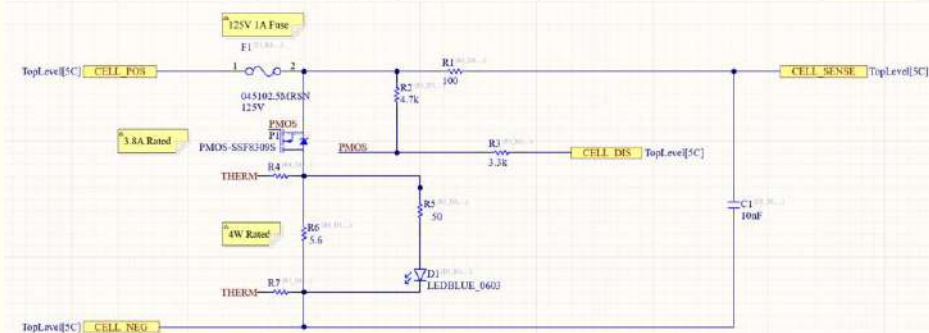
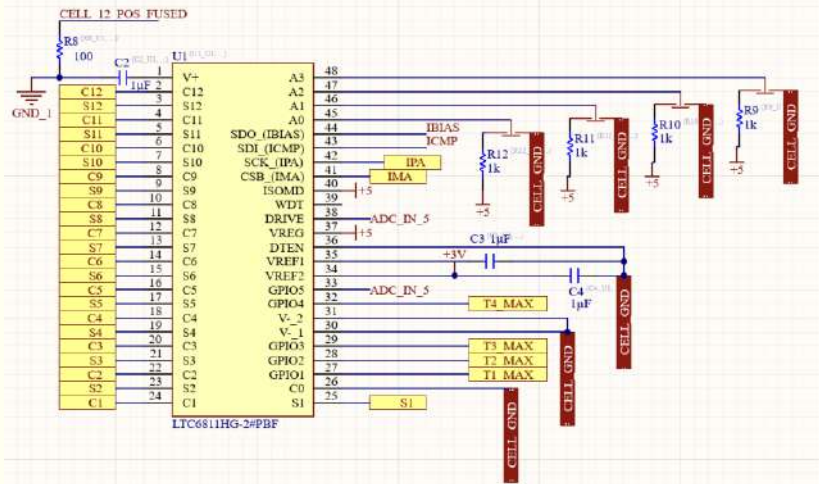
BMS Secondary PCB - Schematics

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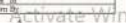


BMS Secondary PCB - Schematics

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Low Power Dog Collar (Collie)

24

Designed **1" x 2"**, **6-layer PCB** integrating mmWave radar, Wi-Fi, GPS, and IMU, with ultra-low 87 mW power draw for dog behavior tracking

Layout focused on digital/analog layout partitioning, impedance controlled traces, and ground stitching for **mixed-signal integrity**

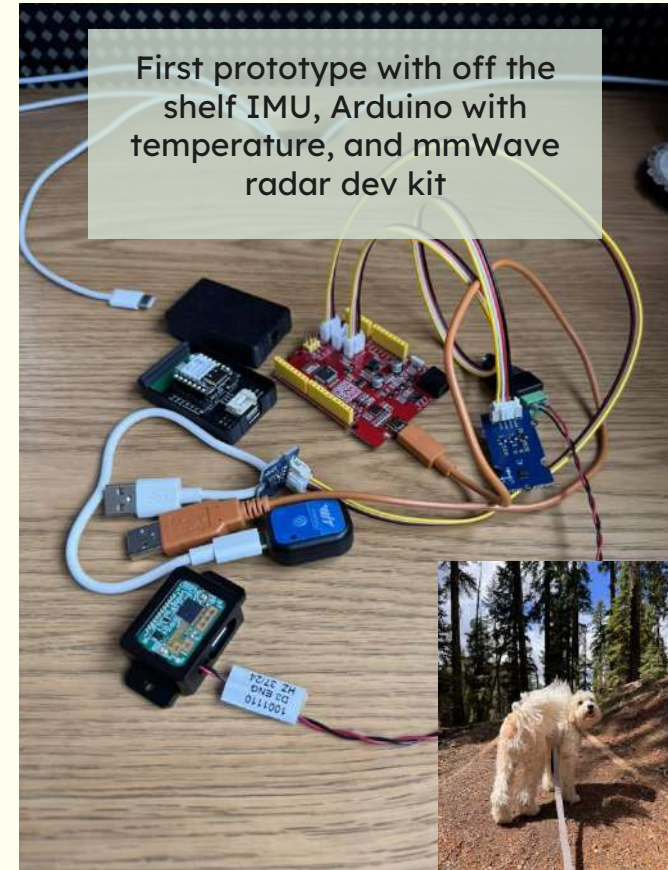
Built V0 without radar module and **designed V1** with radar

Tuned radar detection thresholds with **TI eval board**, validating 91% algorithm accuracy against PPG ground-truth

IMU over I2C, TI mmWave radar over UART, USB-C interface integrated with STM32, BMS for LiPo charging and buck regulation, ESD protection

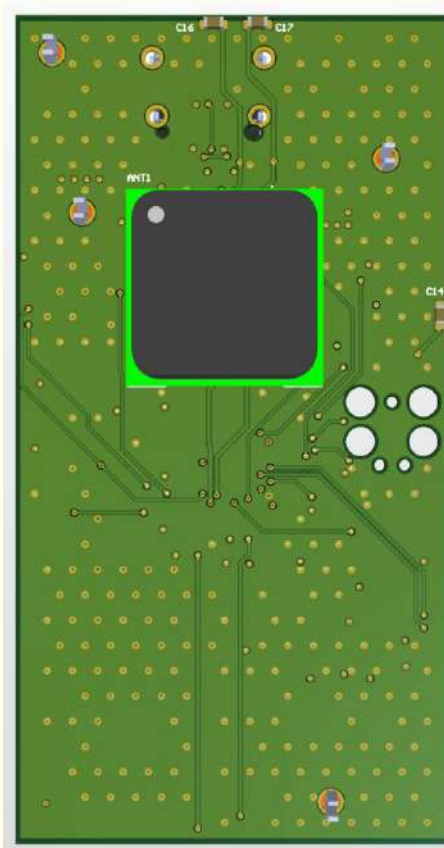
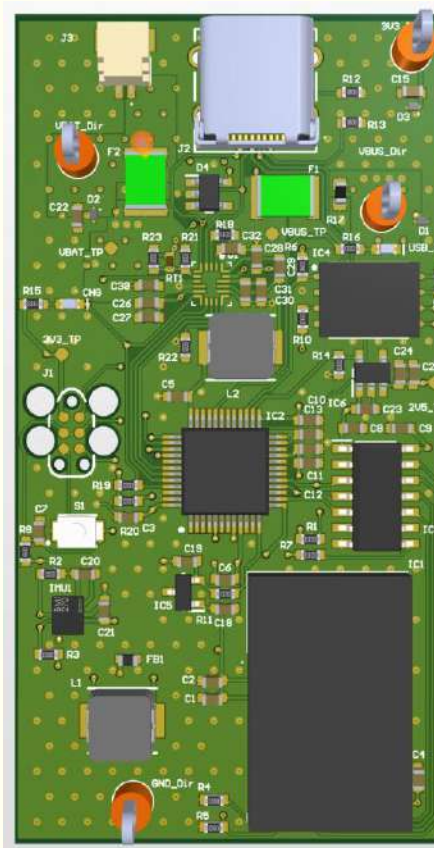
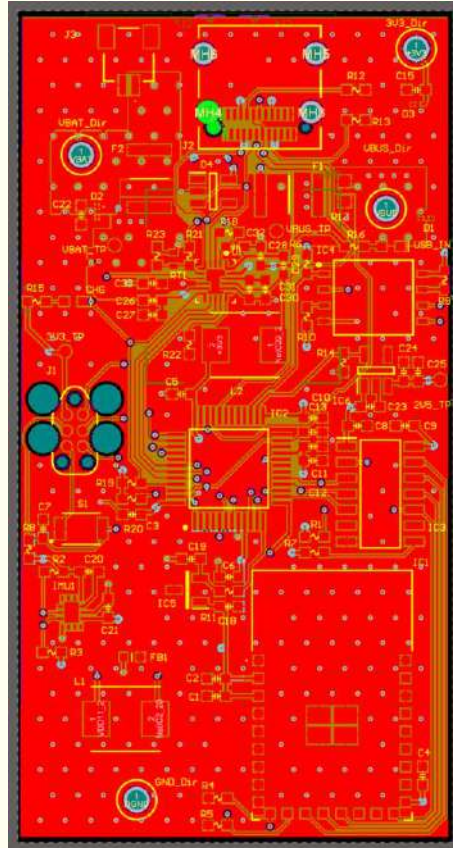
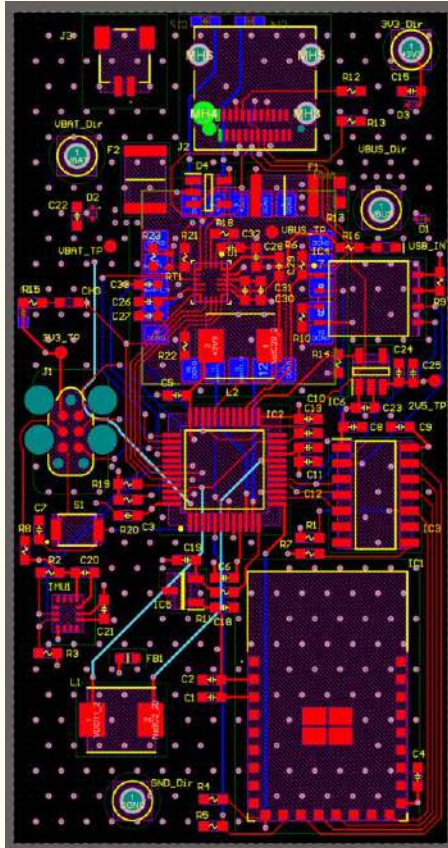
Partnered with Purdue University College of Veterinary Medicine for **planned pilot** and validation program and faculty mentoring

Raised **10k angel funding** to cover prototype development costs, won free incorporation from BuildPurdue startup challenge



V0 Motion Tracking Collar PCB

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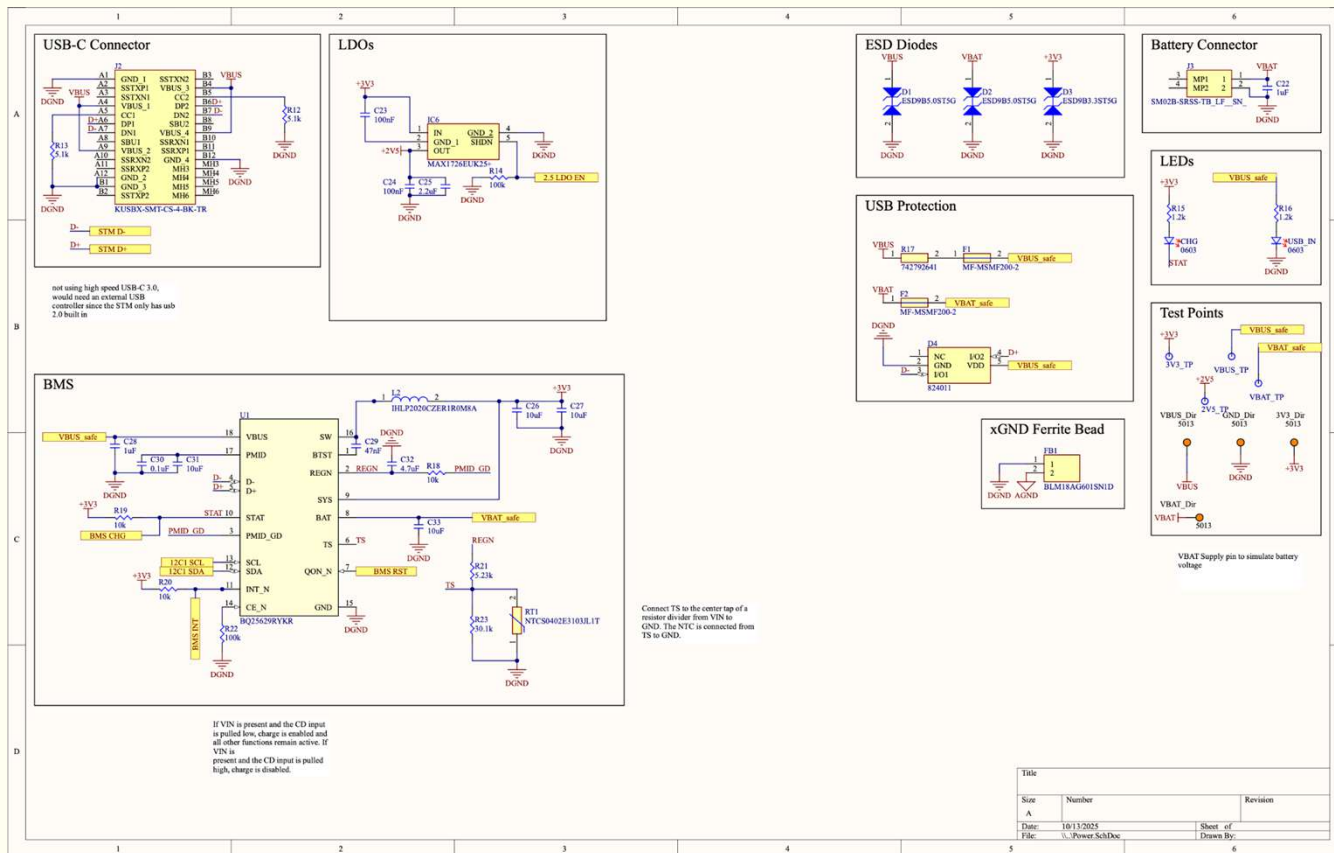


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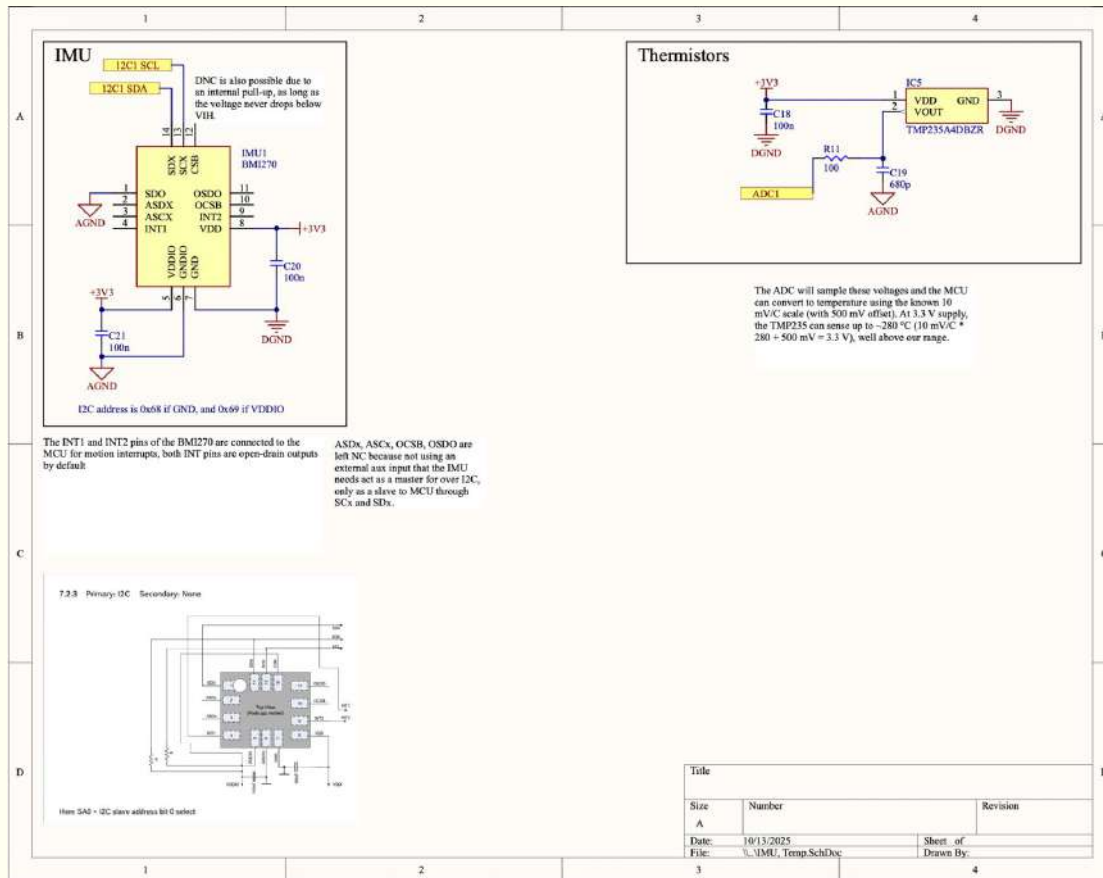
V0 Motion Tracking Collar PCB

27



V0 Motion Tracking Collar PCB

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V0 Motion Tracking Collar PCB

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Fully soldered prototype
with ST-Link for debug
and flashing

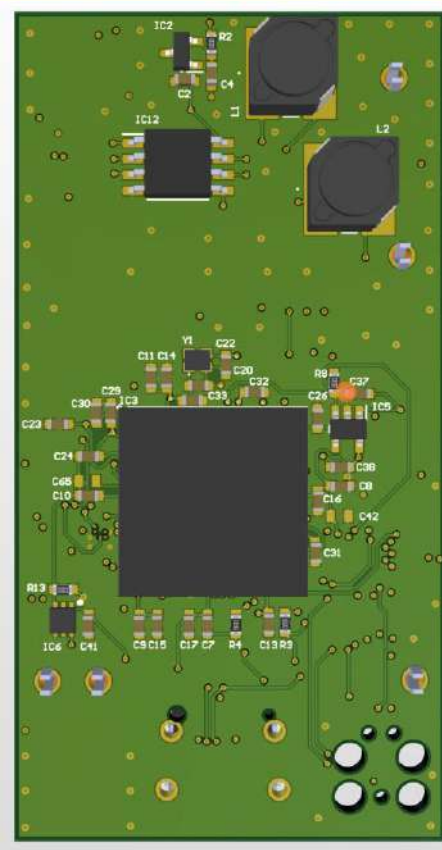
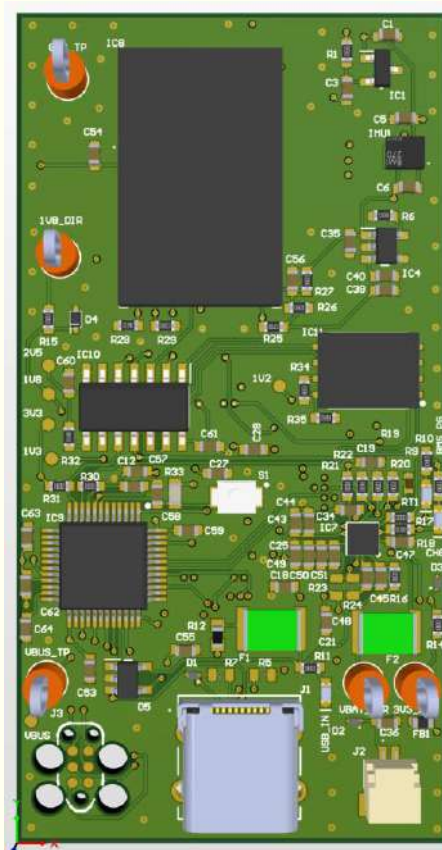
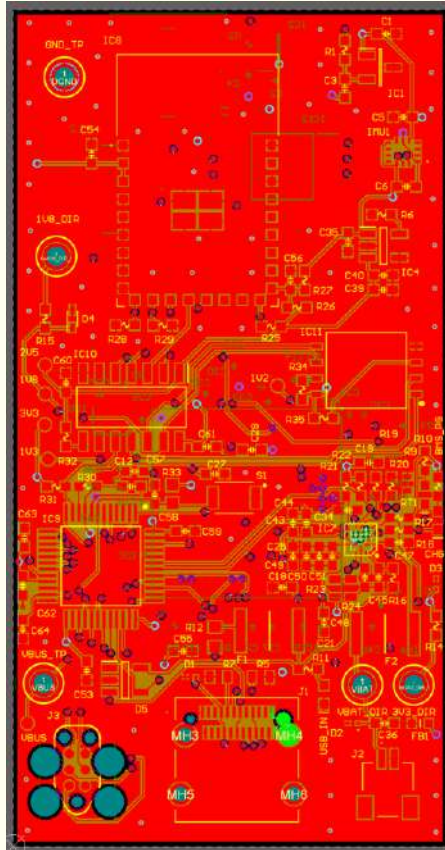
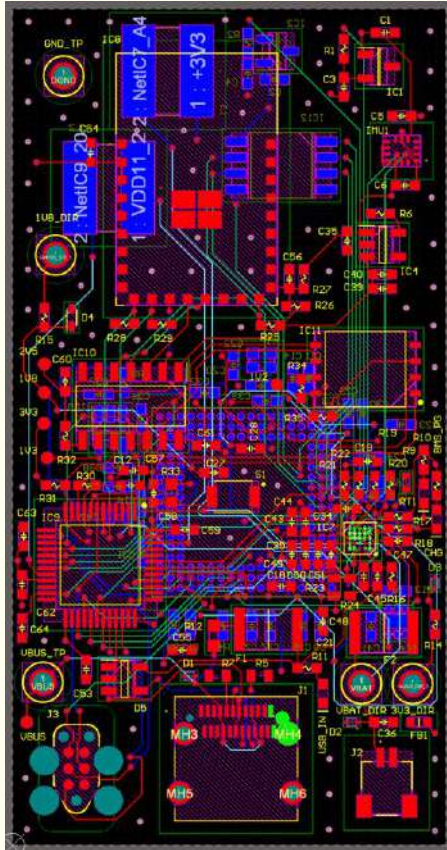


Debugging STM
soldering under
microscope



V1 Radar Heart Rate Collar PCB

30

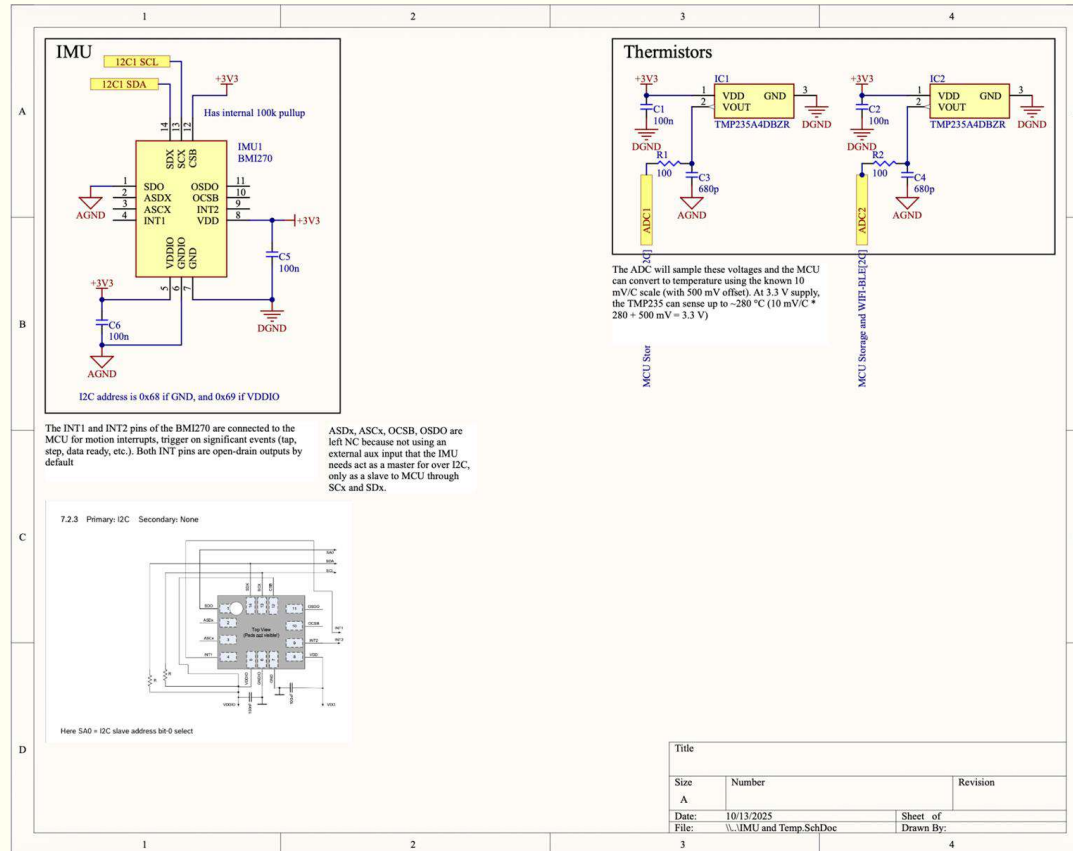


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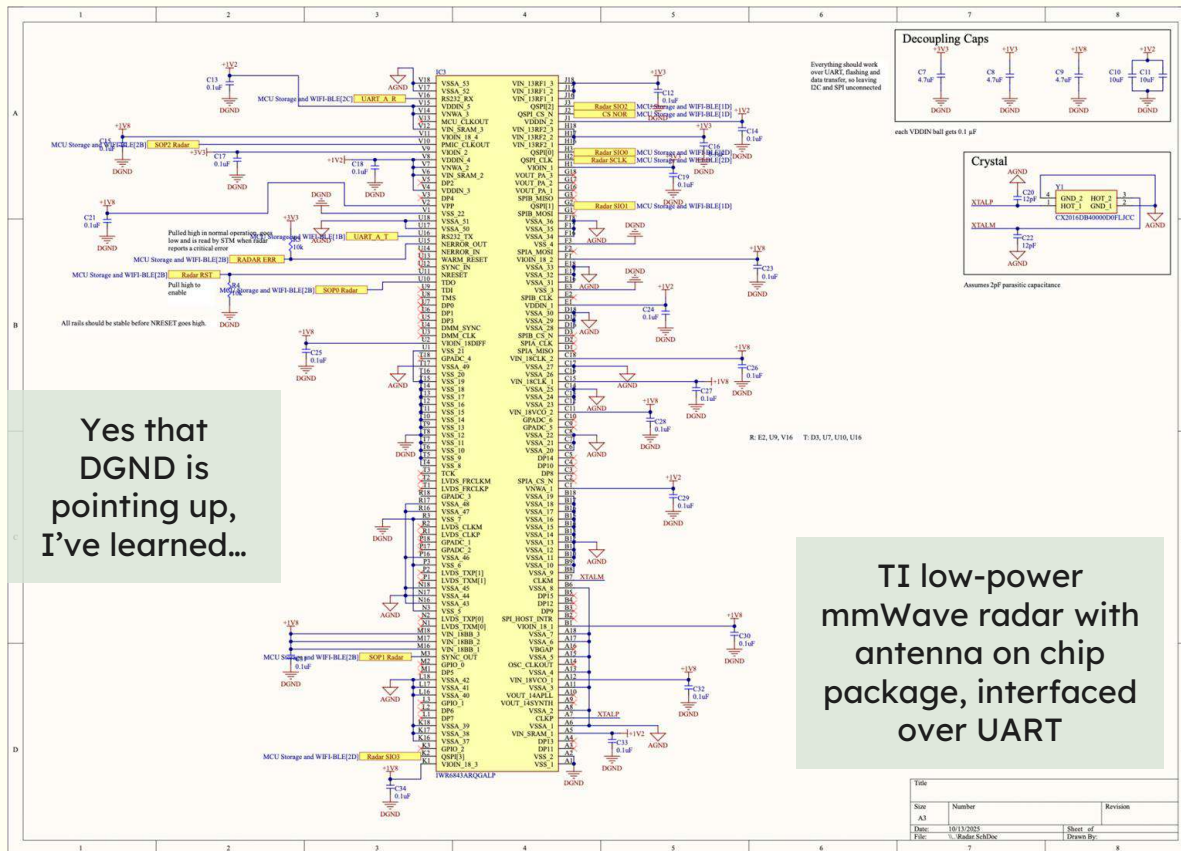
V1 Radar Heart Rate Collar PCB

32



V1 Radar Heart Rate Collar PCB

33



V1 Radar Heart Rate Collar PCB

34

